Summary in Graph

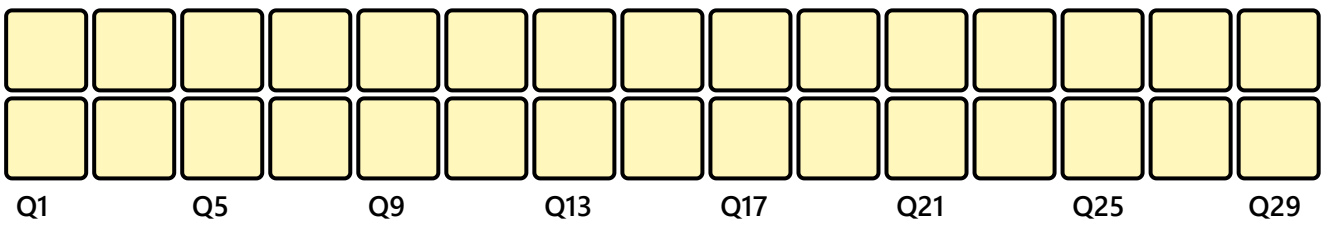
Exam Summary (GO Classes CS Test Series 2026 | C programming | Subject Wise Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (122) out of 1215 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Programming in C

Consider the following recursive function -

```
mystery(a,b){
    if (a < 0 or b < 0) return 0;
    else if (a == 0) return b+1;
    else if (b == 0) return mystery(a-1,1);
    else return mystery(a-1, mystery(a,b-1));
}
```

What will be the value of mystery(3,3)?

Your Answer: | Correct Answer: 61 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What is the content of array A after executing the following code snippet?

```
int A[3] = {1, 2, 3};
int *p;
int **q;
p = A;
p++;
q = &p;
p++;
(*p) = (**q)*2;
```

A. 1 2 3
B. 1 2 4
C. 1 2 6
D. 1 4 3

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be printed by following the C program?

```
#include<stdio.h>
struct Point
{
    int x ;
    int y ;
};

struct Point foo ( struct Point p1 , struct Point * p2 )
{
    p1.x += p2→ x ;
    p2→ y += p1.y;
    return p1 ;
}

int main (){
    struct Point a = {1 , 3};
    struct Point b = {5 , 4};
    struct Point c = foo (a , &b );
    printf ( "%d %d" , c.x , c.y );

}
```

A. 1 3
B. 5 7
C. 6 3

D. 6 6

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

Consider the following lines of C code

```
int a[10]={9,8,7,6,5,4,3,2,1,0};
int i=*(a+4);
int j=&a[3]-&a[1];
int k=*(a+*(a+6));
int m=(&a[5]-a);
```

Which of the following is/are true on the execution of the above lines?

A. Value of i is 5.
B. Value of j is 2.
C. Value of k is 6.
D. Value of m is 5.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Let S be the statement:

```
for i:=1 to N do V[i]:=V[i]+1
```

Which of the following perform(s) the same changes to V as S?

I.
```
i:=0;
while i<=N do
begin i:=i+1 ; V[i]:=V[i]+1 end
```

II.
```
i:=1;
while i<N do
begin V[i]:=V[i]+1 ; i:=i+1 end
```

III.
```
i:=0;
while i<N do
begin V[i+1]:=V[i+1]+1 ; i:=i+1 end
```

A. I only
B. II only
C. III only
D. II and III only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Programming in C

Consider the following C-program:

```
void Demo(int n)
{
    if (n<=1)
    {
        printf("%d", n);
    }
    else
    {
        Demo(n/2);
        printf("%d", n%2);
    }
}
```

Number of 1's in the output if the function is invoked as Demo(220) is _____

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Consider the following function.

```
int f()
{
    int k, result;
    result = 0;
    for ( k = 0; k < 5; k++ )
    {
        if ( ( k % 3 ) == 1 )
            result = result + k;
        else
            result = result + 1;
    }
    return result;
}
```

What value is returned as a result of the call f()?

A. 5
B. 6
C. 7
D. 8

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the output of running file main.c?

```
write.c
**********************
extern int count;

void write_extern()
{
    count +=2;
}
```

```
main.c
**********************
#include "write.c"
#include<stdio.h>
int count = 5;

main()
{
    write_extern();
    write_extern();
    printf("%d", count);
}
```

A. 0
B. 5
C. 9
D. None of these

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Programming in C

What will be output printed by $\text{mystery2}(6)$?

```
void mystery2(int n) {
    if (n > 0) {
        printf("%d", n);
        mystery2(n-2);
        mystery2(n-3);
        printf("%d", n);
    }
}
```

Your Answer: | Correct Answer: 642211431136 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Programming in C

What will be the output printed by $\text{mystery3}(6)$?

```
void mystery3(int n) {
    if (n == 0 || n == 1) return;
    mystery3(n-2);
    printf("%d", n);
    mystery3(n-1);
}
```

Your Answer: | Correct Answer: 243263252432 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

We declare the following variables of different types:

```
int *A1[3][5]
int (*A2)[3][5]
int *(A3[3][5])
int (*A4[3])[5]
```

Let the integer size be 4 bytes and the pointer size is 8 bytes.

Which of the following option(s) is/are CORRECT?

A. sizeof(*A1) $= 40$

B. sizeof(*A2) $= 60$

C. sizeof(*A3) $= 40$

D. sizeof(*A4) $= 8$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12

Numerical Type | Award: 2 | Penalty: 0 | Programming in C

Consider the mutually recursive C functions.

```
int f(int x){
    if (x==0) return 1 ;
        return f(x-1)+g(x-1);
}
int g(int x){
    if (x==0) return 2;
        return g(x-1)+f(x-1);
}
```

What does f(g(2)) evaluate to?

Your Answer: | Correct Answer: 96 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be output if the following function is executed with x equal to 3?

```
void mystery (x)
{
    if (x !=0)
    {
        mystery (x-1);
        mystery (x-1);
        printf("%d", x);
    }
}
```

A. 112211223
B. 3211211
C. 1121123
D. 1121132

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14

Numerical Type | Award: 2 | Penalty: 0 | Programming in C

Consider the following C program.

```c
#include<stdio.h>
int g(int i){
    static int Y = 0;
    Y = Y + i;
    return Y;
}
int f(int i){
    static int X = 0;
    X = X + i;
    X = g(X);
    return X;
}
int main(){
    int i, j;
    for (i = 0; i <= 6; i++)
        j = f(i);
    printf("%d\n", j);
    return 0;
}
```

The value printed by the above program is _____

Your Answer: | Correct Answer: 120 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15

Numerical Type | Award: 2 | Penalty: 0 | Programming in C

Consider the following C-program.

```c
#include<stdio.h>
int Count(int x, int y)
{
    if(x < y) return 0;
    else if(x==y) return x + Count(x-1,y);
    else return y + Count(x-2,y-1);
}
int main()
{
    printf("%d" , Count(9,6));
    return 0;
}
```

The output of the program is _____

Your Answer: | Correct Answer: 18 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16

Numerical Type | Award: 2 | Penalty: 0 | Programming in C

What will be printed by the following program?

```c
#include<stdio.h>
int func(int n, int * fg) {
    int t, f;
    if (n <= 1) {
        *fg = 1;
        return 1;
    }
    t = func(n - 1, fg);
    f = t + *fg;
    *fg = t;
    return f;
}
int main() {
    int x = 15;
    printf("%d\n", func(5, &x));
}
```

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider the following C code where main shows the intended use of the functions relating to struct foo.

```c
struct foo {
    int x;
    int y;
    int * p;
};

struct foo* make_foo() {
    struct foo *foo = malloc(sizeof(struct foo ));
    foo->y = 2;
    foo->x = foo->y*2;
    foo->p = malloc(sizeof(int));
    *(foo->p) = 42;
    return foo;
}

void update_foo(struct foo *foo) {
    int num = *(foo->p);
    foo->p = malloc(sizeof(int));
    *(foo->p) = num + 1;
}

void free_foo(struct foo *foo) {
    free(foo);
    free(foo->p);
}

int main() {
    struct foo *f = make_foo();
    update_foo(f);
    free_foo(f);
}
```

Which of the following is true about the given program?

A. Given program has a memory leak
B. Given program has a dangling pointer
C. Given program may result in run time error
D. Given program will result in a compile-time error

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

What does the following program print?

```
#include<stdio.h>
#include<string.h>
struct MdbRec {
    char name[16];
    char msg[24];
};
int main()
{
    struct MdbRec m[1];
    strcpy(m[0].name, "dude");
    strcpy(m[0].msg, "0");

    if(m == m->name)
        printf("%d", m->msg - (char *)m );
}
```

A. Prints nothing
B. Prints 17
C. Prints 24
D. Prints 16

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Mark all options which are likely to create problems with memory, i.e., run time error or have a memory leak. Assume malloc is always successful, and sizes are as follows -

```
sizeof(int *) = 8 bytes
sizeof(double) = 8 bytes
sizeof(char) = 1 byte
sizeof(int) = 4 byte
```

```
Program 1
int **p;
p = (int **)malloc( 4 * sizeof(int) );
for (int i = 0; i < 4; i++) {
    p[i] = (int *)malloc( sizeof(int) );
}
```

```
Program 2
int *p = malloc(sizeof(double));
*p = 0;
free(p);
```

```
Program 3
struct p { double x; double y; };
struct p *array = malloc(n * sizeof(struct p *));
for (int i=0; i<n; i+=1) array[i].x=2.3
```

```
Program 4
int *p = malloc(sizeof(char));
*p = 0;
```

A. Program 1
B. Program 2
C. Program 3
D. Program 4

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider two files given below.

```
//file1.c
#include<stdio.h>
main ()
{
  extern int i;
  printf("file1");
}
```

```
//file2.c
#include<stdio.h>
main ()
{
  int i =1;
  printf("file2");
}
```

We compile both files independently. And link them in case compilation is successful.

Which of the following is/are TRUE ?

A. file1.c and file2.c can be compiled independently and linked together
B. file1.c has a compilation error since extern is a local variable
C. file1.c and file2.c can be compiled independently but can not be linked together
D. file1.c can be compiled and linked independently of file2.c.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be the output of the following program?

```
#include<stdio.h>
int rec(int x, int y) {
static int count = 0;
if (x == 0)
  return count;
count++;
if (x > y)
  rec(x - y, y);
else
  rec(x, y - x);
return count;
}
main() {
    int i = 10, j = 2, n;
    n = rec(i, j);
    printf("%d", n);
}
```

A. 4

B. 5
C. 6
D. Infinite loop

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be the output of the program below-

```c
int i = 1;
void my_extern1(void);
void my_extern2(void);

int main() {
    int count = 0;

    while (i++<5) {
        static int i = 3;
        i++;
        count +=i;
        my_extern1();
    }
    printf("%d", count);
}

void my_extern1(){
    extern int i;
    if(i++ > 0) my_extern2();
}

void my_extern2(){
    extern int i;
    if (i++ < 3) my_extern1();
}
```

A. Infinite recursion
B. 9
C. 4
D. 5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider the k bit binary pattern on $2's$ complement system.

- $\mathrm{T}_{\max}$ and $\mathrm{T}_{\min}$ are maximum and minimum signed numbers we can represent using k bits.
- $\mathrm{U}_{\max}$ and $\mathrm{U}_{\min}$ are maximum and minimum unsigned numbers we can represent using k bits.

Which of the following(s) is/are true for $k = 32$?

A. $\mathrm{T}_{\min} = 1 << 31$
B. $\mathrm{T}_{\max} = \sim (1 >> 31)$
C. $\mathrm{U}_{\min} =!(1 >> 31)$
D. $\mathrm{U}_{\max} = \sim (1 >> 31)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What is the output of the following code? Assume that int is 32 bits, short is 16 bits, and the representation is two's complement.

```
unsigned int x = 0xDEADBEEF;
unsigned short y = 0xFFFF;
signed int z = -1;
if (x > (signed short) y)
    printf("Hello");
if (x > z)
    printf("World");
```

A. Prints nothing
B. Prints "Hello"
C. Prints "World"
D. Prints "HelloWorld"

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Numerical Type | Award: 2 | Penalty: 0 | Programming in C

What will be the output of the following C program?

Assume that negative numbers are represented using $2's$ complement.

```
#include<stdio.h>
void main()
{
    int a = 2;
    a = ~a+2 << 1;
    printf("%d",a);
}
```

Your Answer: | Correct Answer: -2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Numerical Type | Award: 2 | Penalty: 0 | Programming in C

What will be the output of the following program?

```
#include<stdio.h>
main()
{
    int a[2][3][4];
    printf("%d", a[1][0]-a[0][2]);
}
```

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

```c
#include<stdio.h>
int bar(int data[], int i, int a, int x) {
    if (x<0) return 0;
    x--;
    int b = a + data[i];
    if(data[i] >0) return bar(data, i+1, b, x);
    if(data[i] == 0 || i ==x-1) return a;
    return -a;
}
int main()
{
    int data[5] = {1,2,3, -1};
    int x = (int *)(&data+1)-data;
    printf("%d", bar(data, 0, 0, x) );
}
```

What will be the output of a given program?

A. 6
B. -6
C. 0
D. 5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #28 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Which of the following expressions will evaluate to true?
INT_MIN is the minimum signed integer in the system which is using $2's$ complement representation for signed integers and variables are defined as follows:

```c
int x = 5, i = -7;
```

A. ((x >> 2) << 2) <= x
B. ~ 1&& 1
C. INT_MIN == -INT_MIN
D. -10 < i <-1

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Numerical Type | Award: 2 | Penalty: 0 | Programming in C

How many times is foo activated (called), including the first "$\text{foo}(3, 12)$"

$\max()$ and $\min()$ are functions that return maximum and minimum respectively.

```c
int foo(int a, int b) {
    if (a==b) {
        return b;
    }
    int mn =min(a,b), mx =max(a,b);
        return foo(mn,mn) + foo( mx - mn , mn);
}
```

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Programming in C

Consider the following recursive function. What is $f(0)$?

```
int f(int x) {
    if (x > 1000) return x - 4;
    else return f(f(x+5));
}
```

Your Answer: | Correct Answer: 997 | Not Attempted | Time taken: 00min 00sec | Discuss

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